Why-Ledger

An agent that checks every claim against the data

Track 2: The Agent That Can Explain Why | Open Agent Hackathon 2026

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AI-assisted: production code written by Cursor (cursor-agent), directed and reviewed with Claude Code; Narration: AI text-to-speech voice (Kokoro).

Problem and target users

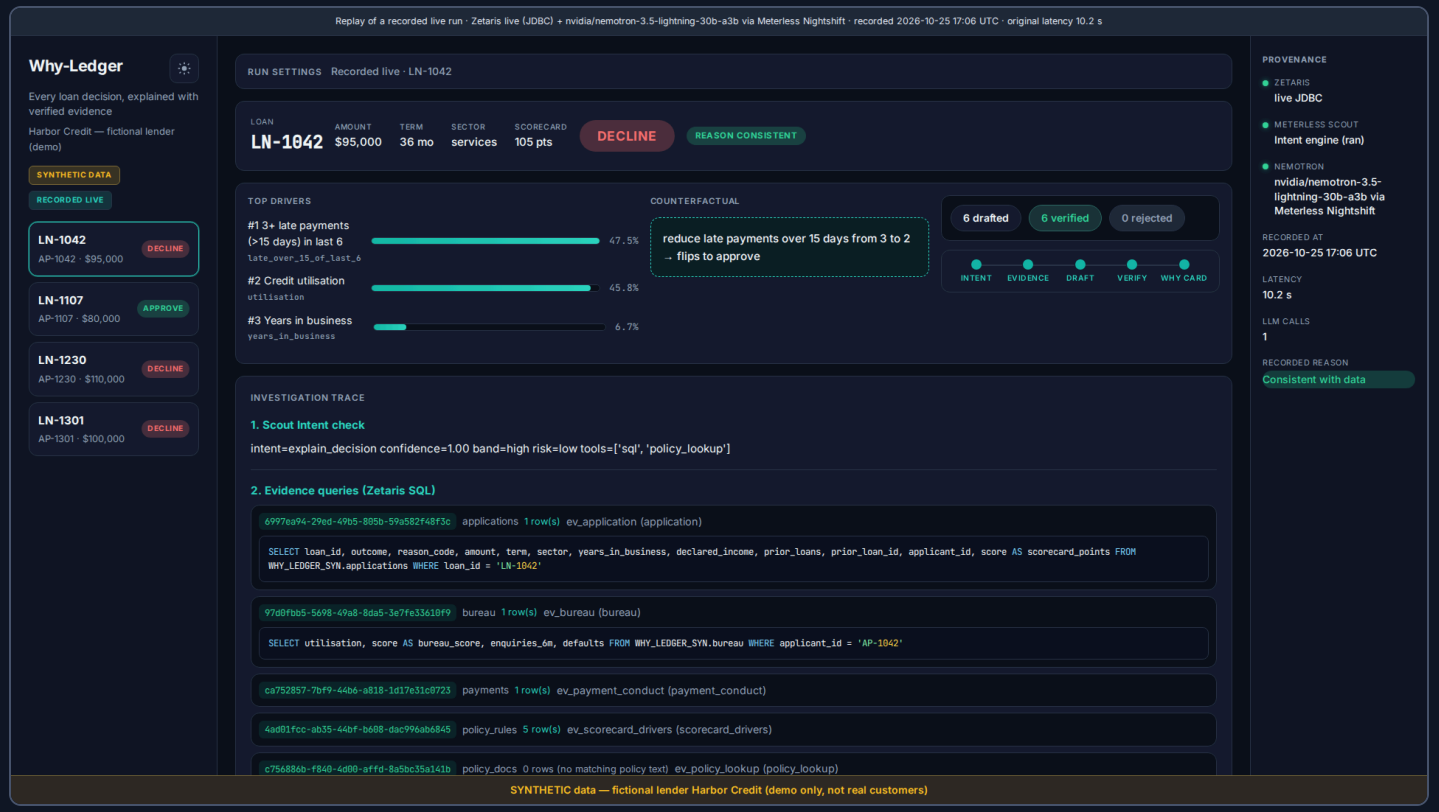
The problem

- A loan decision depends on applications, payments, bureau data and policy rules held in different systems.
- Regulators and applicants need the reasons.
- An LLM explanation can state a value or a reason that the data does not support.
- Even the reason code stored on the decision record can be wrong.

Who it is for

- Credit analysts and reviewers
- Model-risk and compliance teams
- Applicants, through an illustrative reasons letter
- No time-saved or accuracy figures are claimed; all data is synthetic.

The solution: a Why-Card in which every claim is checked

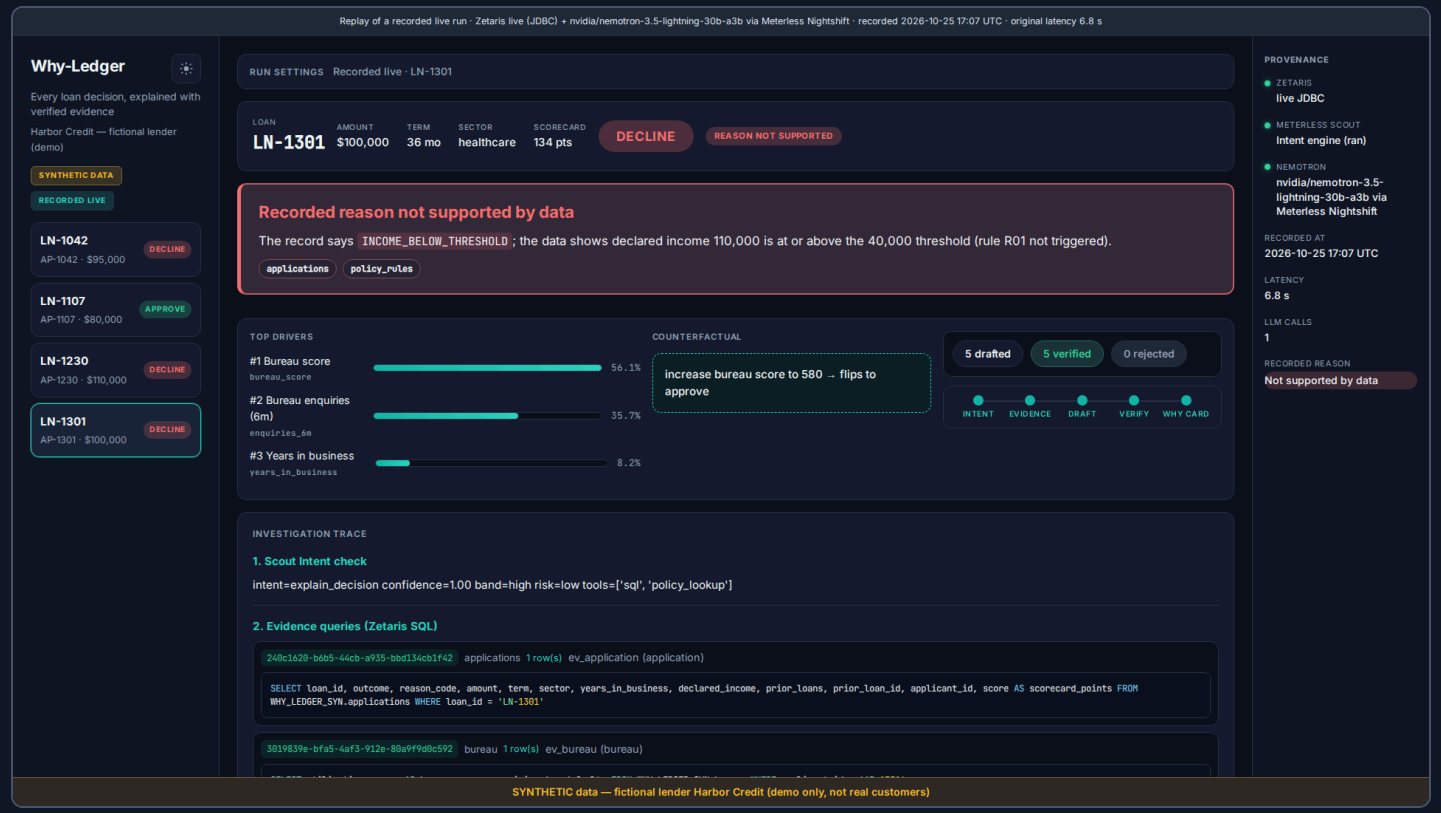


LN-1042 (recorded live run, SYNTHETIC): drivers, counterfactual, 6 drafted / 6 verified / 0 rejected

- Drivers ranked from scorecard points in the data, not by the model
- Only VERIFIED claims reach the card; unsupported drafts go to a rejected panel
- Counterfactual for declines: the smallest change that flips the decision
- Evidence drawer: SQL, rows and SHA-256 for every claim
- Illustrative applicant letter of ranked principal reasons (not legal advice)

It caught the lender's own wrong reason

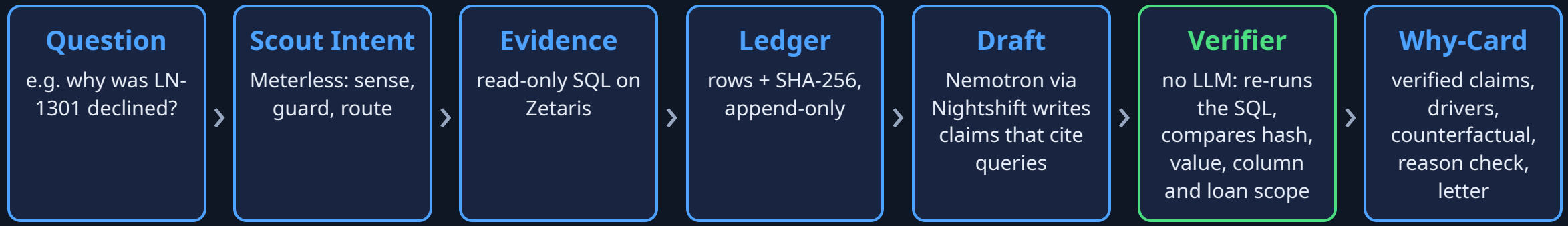
Loan LN-1301



LN-1301: red flag, Recorded reason not supported by data (SYNTHETIC)

- The decision record says INCOME_BELOW_THRESHOLD. Declared income is 110,000, above the 40,000 policy threshold, so rule R01 is not triggered.
- The check re-runs SQL on Zetaris and shows record and data side by side. It states what the record says; it does not assign blame.
- Data-supported drivers: bureau score 56.1 %, credit enquiries 35.7 %. Counterfactual: bureau score 580 flips the decline to approve.
- Planted mismatches flagged: 1 of 1. False flags on the other 3 loans: 0.

Agent workflow and architecture

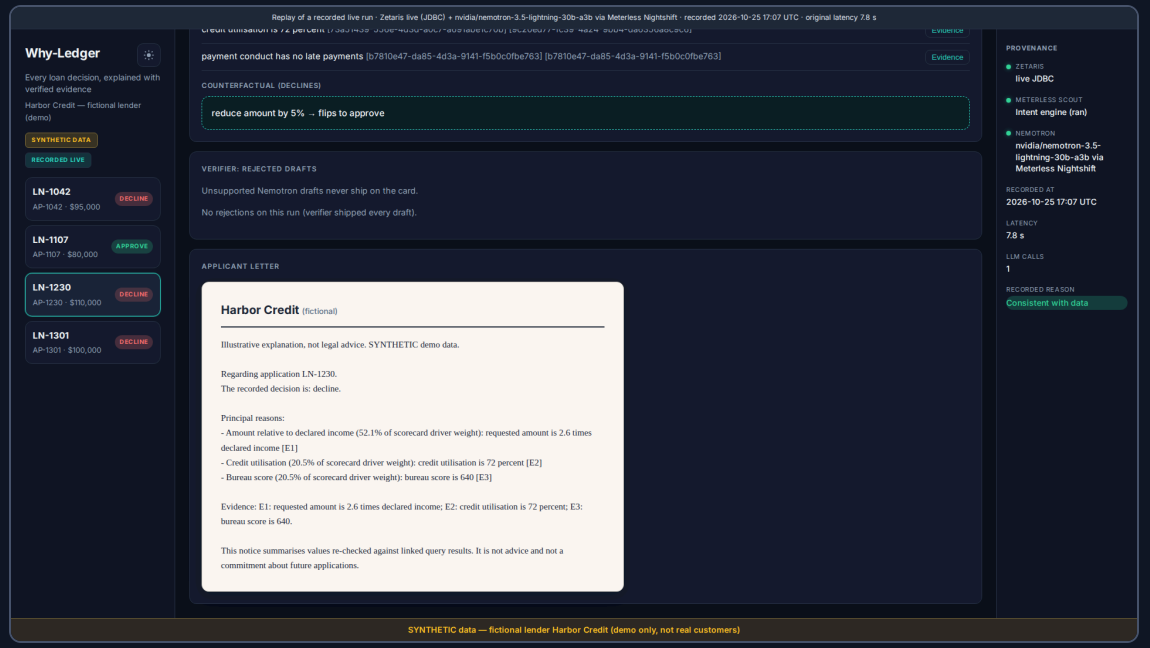


- Zetaris is the only data path; the model sees data only through logged queries.
- Planned mode: code chooses the evidence queries and one Nemotron call drafts the claims. Rejected claims are shown, never hidden.
- Recorded-reason check (deterministic): compares the reason code on the record with the policy rule and the data.

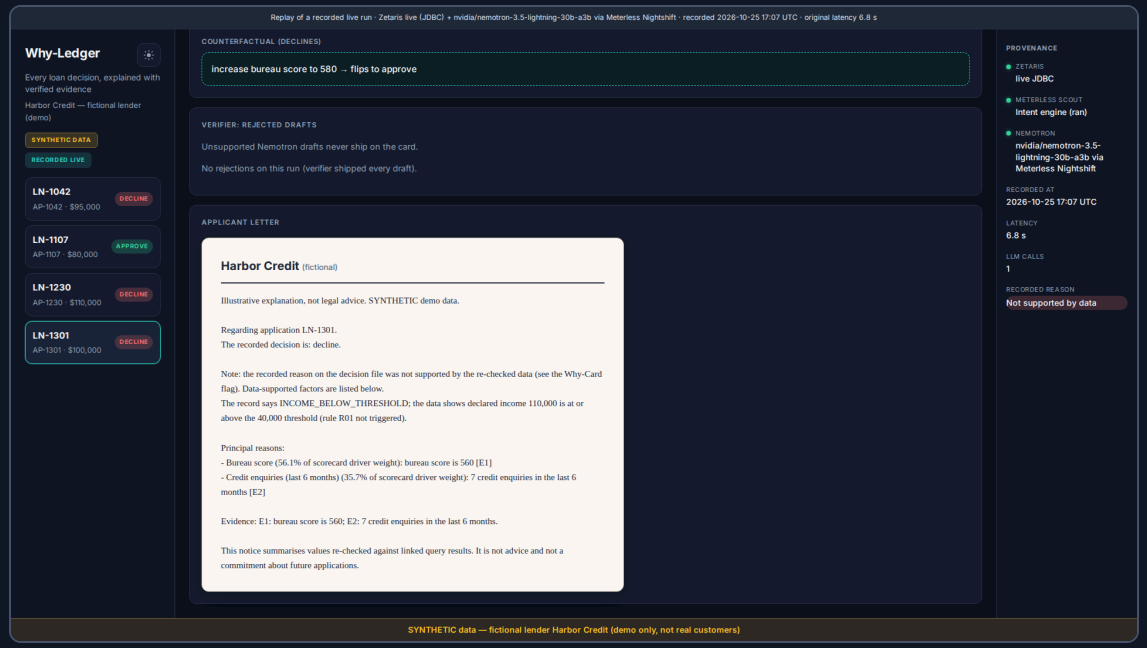
Technical stack and sponsor technology

Zetaris	Sole data path. Live JDBC to Zetaris 2.4.3.1 (local Docker) over SYNTHETIC tables; agent queries and verifier re-runs are read-only SQL. It is also the verifier's oracle for what the model cited.
Meterless Scout Intent	Every question is sensed, guarded and routed before any model runs; injected prompts are refused (tested on a port of the upstream adversarial set).
Meterless Nightshift	Primary LLM provider: an OpenAI-compatible runtime on the Windows host, reached from WSL. All four recorded runs show provider meterless-nightshift.
NVIDIA Nemotron	nvidia/nemotron-3.5-lightning-30b-a3b via Nightshift drafts the claims (1 call per loan).
Cursor	All production code (src/, scripts/, UI, tests) written by cursor-agent; every task logged in docs/CURSOR-LOG.md.
What is unique	The model's own cited SQL is re-executed on Zetaris and each stated value is tied to its named column; the record's reason code is checked against the data too.

Product visuals: evidence-backed applicant letter



LN-1230: principal reasons in rank order, #1 amount-to-income 52.1 %



LN-1301: notes the unsupported recorded reason, lists data-supported reasons

Built from logged, hash-checked query evidence ([E1], [E2] chips point to query results). Illustrative, not legal advice.

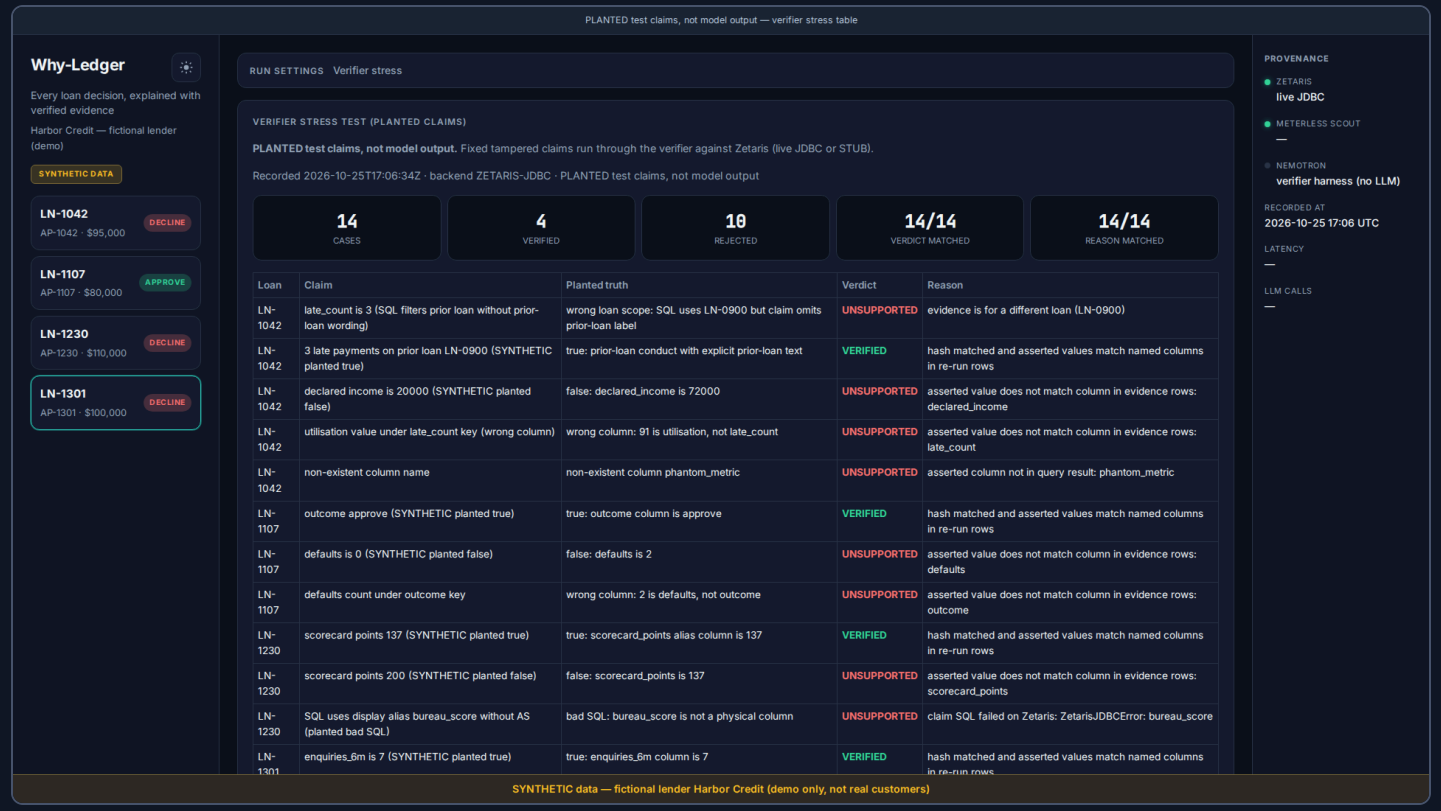
Evaluation: recorded Nightshift run, 25 Oct 2026

eval/results-live-nightshift.md and replay bundles (reason check): four planted loans, one run each, SYNTHETIC

Loan	Drafted	Verified	Rejected	Top driver = truth	Latency
LN-1042	6	6	0	yes	10.2 s
LN-1107	5	5	0	yes	6.7 s
LN-1230	6	6	0	yes	7.8 s
LN-1301	5	5	0	yes	6.8 s
Total	**22**	**22**	**0**	**4/4**	

- Recorded-reason check (from the replay bundles): 1 of 1 planted mismatch flagged (LN-1301); 0 false flags on the other 3.
- Scout intent routing 12/12 on 12 author-written cases (eval/results-live-zetaris.md). One run per loan: no variance claim.
- An earlier recording the same day took 67 s to 534 s on the shared free-tier model; this one 6.7 s to 10.2 s.

Verifier stress test and honest limits



PLANTED test claims, not model output

- 14 planted cases on live Zetaris: 4 VERIFIED, 10 UNSUPPORTED
- Verdict matched 14/14; intended failure reason matched 14/14 (reported separately)
- Honest failure: an earlier LN-1230 attempt crashed on a non-existent column; the verifier now rejects such claims
- Limits: synthetic data, 4 loans, one Nemotron model works on this account, one run per loan, not compliance advice

Future enhancements

- Model-proposed SQL loop mode with a hardened verifier (the one live loop run was weak: eval/live-nemotron-smoke.md)
- Federation across more Zetaris sources inside one verified query
- Richer policy engine and more reason-code checks beyond income
- Meterless H-MEM precedent memory: reuse verified explanations across similar cases
- Multiple runs per loan for latency and agreement ranges
- Real-data governance review before any production use

Team and disclosure

- Entrant: Harsha Shinde (solo).
- Production code written by Cursor (cursor-agent; composer-2.5, Cursor Auto for the first tasks); log in docs/CURSOR-LOG.md.
- Directed and reviewed with Claude Code (Anthropic): design, prompts, docs, test review, video and deck assembly.
- Narration: AI text-to-speech voice (Kokoro).
- All data, Harbor Credit, applicants and loan IDs are fictional and labelled SYNTHETIC.
- Repo started 22 Oct 2026; no pre-existing components. Numbers come from recorded runs in eval/.
- Reason-check and letter fields in the recorded bundles were regenerated offline, deterministically, from the stored evidence rows (no new model call).
- Repo: https://github.com/xexpertai/why-ledger